

Investigation**Woodman's Point**
Calculator allowedName: Mr Wilson

Total Marks: 17 marks

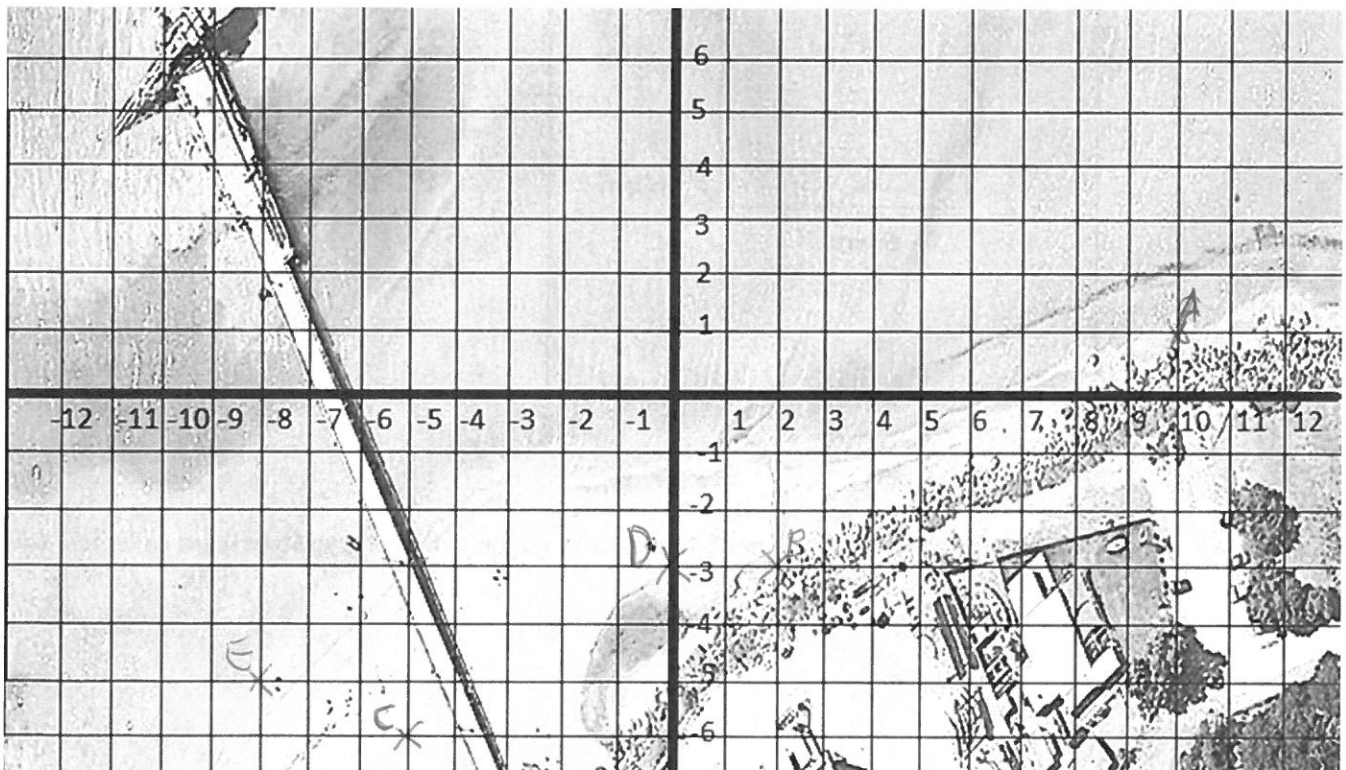
Total Time: 20 minutes

A group of students are conducting an experiment to determine what bins are required at Woodman's point.

1. (5 marks)

The students recorded the location of where they found the following items at Woodman's point. Using the location coordinates from the table, plot on the map where each of the items of rubbish was found.

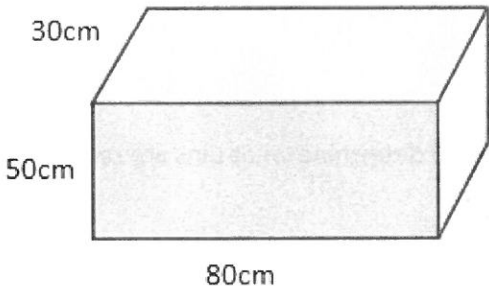
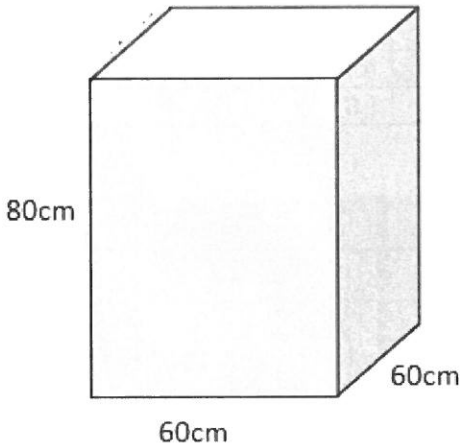
Item	Location (x,y)
a) Glass bottles	(10, 1)
b) Paper bags	(2, -3)
c) Food scraps	(-5, -6)
d) Aluminium Cans	(0, -3)
e) Plastic bags	(-8, -5)

✓
✓
✓
✓
✓*1 mark each*

2. (8 marks)

The rubbish collected was placed in to the following containers.

Complete the following table. (Hint: 1L = 1000 cm³)

Type of Material	Container	Volume (cm ³)	Capacity (L)
Recyclables		$30 \times 50 \times 80 \text{ cm}^3$ $= 120,000 \text{ cm}^3$	$120000 \div 1000$ $= 120 \text{ L}$
General waste		$80 \times 60 \times 60$ $= 288,000 \text{ cm}^3$	$288,000 \div 1000$ $= 288 \text{ L}$

- a) If they filled the Recyclables container 3 times, estimate how much recyclables were collected. Give your answer in litres.

$$120 \text{ L} \times 3 = 360 \text{ L}$$

- b) If they filled the General waste container 5 times, estimate how much general waste was collected. Give your answer in Litres.

$$288 \text{ L} \times 5 = 1440 \text{ L}$$

3. (4 marks)

The bins that they plan on using for Woodman's point have a capacity of 240L. If both recycling and general waste bins are the same size determine;

a. How many recycling bins would be required to dispose of the recyclable rubbish?

$$\frac{360L}{240L} = 1.5 \text{ bins. } \checkmark$$

$\therefore$ we would need 2 bins. $\checkmark$

b. How many general waste bins would be required to dispose of the general waste rubbish?

$$\frac{1440L}{240L} = 6 \text{ bins. } \checkmark$$

6 bins will be required. $\checkmark$

